

Reducing Greenhouse Gases—and Reaping the Benefits

by Steve Cowell

Conservation Services Group (CSG), the nonprofit that I manage, has been designing, developing, and delivering innovative energy efficiency and renewable energy programs for more than two decades. Starting in the late '90s CSG began exploring a way to reap additional value from the efficiency and energy projects that we implement. We got involved in a relatively new kind of market—the environmental benefits market. This market is similar to the stock market, but instead of trading stocks, what gets bought and sold are environmental benefits—reductions in emissions of pollutants, such as nitrogen oxide (NO_x), carbon dioxide (CO₂), or other greenhouse gases (GHGs).

How does selling an environmental benefit work? Say, for example, that a regulated company is required to cut its NO_x emissions by 10%. If it cuts its NO_x emissions by more than 10%, it can sell the difference, in the form of a NO_x credit or allowance, to other companies that cannot meet the legal limit in a cost-effective manner. In addition to companies selling allowances, some states now permit such trading by efficiency providers, who deliver energy savings that result in lower NO_x emissions.

When considering any environmental trading, it is important to remember that any measure that reduces the consumption of energy, or eliminates the need to burn fossil fuels, reduces pollution from CO₂ and from other GHGs. Methane is a more potent GHG than CO₂ and several other gases are also serious pollutants. For this reason, I prefer to focus on GHGs and

GHG markets, rather than on CO₂ pollution alone.

A market for buying and selling allowances for CO₂ and other GHGs was started in Europe in 2005 as a means for helping European countries to cut GHG emissions. A similar market is just getting started in the United States. In both markets, GHG allowances can be created from reducing emissions by, for example, implementing efficiency improvements or installing on-site renewable energy or combined heat and power (CHP) projects. These allowances can be sold to businesses or even to individuals who want to reduce their personal carbon footprint.

CSG is actively seeking ways and markets to sell the reductions in pollutant emissions created through our efficiency improvements. So far, CSG has traded emissions reductions acquired from renewable energy projects, including solar, wind, and landfill gas; from energy efficiency measures; and from CHP projects. The experience we acquired from selling the benefits stemming from these site-installed projects will help to pave the path for us and other efficiency providers to more broadly sell the environmental benefits that we create in the fledgling carbon and GHG markets. As environmental markets increasingly reflect the benefits that our work delivers, expanded resources will become available to help stimulate

additional energy efficiency work in new and existing homes. Also, I think these markets are essential tools to combat global warming.

A Tale of Two Trades

The first environmental benefits market, the NO_x market, was created by the Environmental Protection Agency (EPA) to help states to achieve compliance with the Clean Air Act. Under the provisions of this act, EPA specifies how much NO_x all entities in a given state can emit. The state is permitted to sell a certain number of NO_x allowances to entities that exceed their limit, if they can't meet their regulated emissions level through the implementation of pollution prevention technologies. States may set aside a percentage of these allowances for energy efficiency projects. In Massachusetts, CSG helped to persuade the Department of Environmental Protection to agree to a 5% set-aside for efficiency. This figure represents 687 tons of NO_x (1 ton = 1 allowance).

To trade in this market, we calculated the reduction in the use of gas and electricity—and the corresponding reduction in NO_x emissions—that our customers received when we implemented energy efficiency projects in new or existing homes. We then aggregated these reductions to create approved allowances to sell. The projects in



An Energy Star home, such as this one in Massachusetts, with its increased insulation, advanced air sealing, Energy Star appliances and lighting, an efficient furnace, and solar electric panels can cut electricity use by more than 2,200 kWh annually and also reduce the associated emissions of carbon dioxide, sulfur oxides, and nitrogen oxides.

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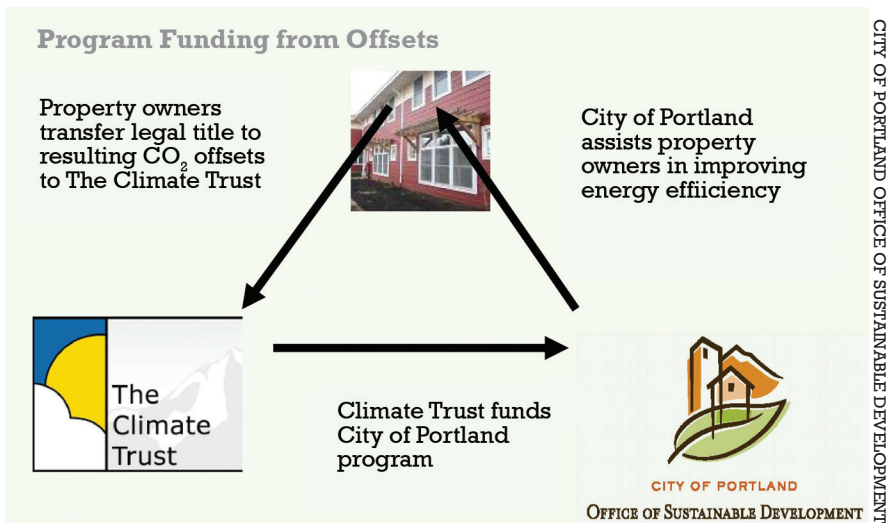


Figure 1. In Portland, the sale of carbon offsets helps to fund energy-efficient home retrofits, which then create more offsets to sell.

question included the implementation of efficiency measures in Energy Star homes, along with the installation of insulation, compact fluorescent light bulbs (CFLs), and other efficiency improvements in 17,000 existing homes in Massachusetts. The energy savings from the installation of these measures were carefully documented by our energy auditors and the resulting NO_x emissions reductions were then calculated using a method that had been approved by the Massachusetts Department of Environmental Protection. The results were submitted to the Massachusetts Department of Environmental Protection, and we were approved to sell the NO_x allowances that we had created. We have now been documenting NO_x emissions reductions and selling the resulting allowances for four years. The amount of NO_x allowances that we have sold has ranged from 25 tons to 40 tons per year. The sale price of each allowance has ranged from \$1,000 to \$2,000 per ton, totaling between \$40,000 and \$80,000 per year. To give a concrete example, each incandescent bulb switched out to a CFL creates about \$0.07 per year in value—that is, it is worth \$0.07 per year on the NO_x market—or about \$0.50 over the life of the bulb. This represents about 25% of the value of the bulb.

On the other side of the continent, we are participating in the nascent carbon market by helping to generate carbon allowances, each of which is

equivalent to 1 ton of carbon emissions avoided. These allowances were created when the Portland Office of Sustainable Development (OSD) and CSG retrofitted multifamily buildings with funding from the Climate Trust of Oregon and the Energy Trust of Oregon (see Figure 1). As a result of our combined efforts, thousands of housing units were repaired, inefficient energy use was cut, and CO₂ emissions were reduced.

Perfecting the Trade

The two examples given above show how environmental benefits can be aggregated to reap additional revenue from energy efficiency investments. Here are some of the lessons that CSG learned from these trading experiences:

- Measurement, verification, and certification of CO₂ and other GHG emissions reductions from implementing energy efficiency projects requires careful and thorough documentation so that any supervising agency can feel confident that the work in question was actually done, and that it produced the environmental benefits claimed. The measurement and verification (M and V) process consists of two steps. The first step is to verify that the measures have been installed—a process that usually requires a review by a third party. The second step is to

quantify the emissions reductions. This step requires sound statistical accuracy.

- Database systems are needed to aggregate large numbers of transactions and the savings that those transactions deliver. This is necessary in order to establish the basis for verification. The calculation of the savings from each installation depends on tracking each specific measure installed at each location. Although the data systems that we have employed have been adequate to verify emissions reductions, we are still working to improve our M and V process. In the case of the Oregon program, the funding source was not entirely satisfied that our M and V process fully proved that the reductions were the result solely of the carbon payments.
- The property owner, the delivery contractor, and the funding source must all agree on exactly who owns the benefits. (Disputes lead to paralysis!) As a result of our trading experiences and dispute negotiations that we have been involved with, CSG has created contract language to clarify this gray area.
- The idea that carbon reductions result from carbon market payments, and not from other benefits, such as utility incentives or cost-justified energy savings, creates a difficult (and in our view unnecessary) hurdle to overcome. It will be necessary to resolve this additionality, or free ridership, issue to realize the full value of the environmental benefits derived from energy efficiency projects.

Ultimately, the environmental benefits of energy efficiency and all clean, customer-sited demand resources will be incorporated into our business proposition.

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